

02425 Diffusions and Stochastic Differential Equations

Exercises Week 1: Diffusive Transport in Shear Flow

Viggo Starcke
s250690@dtu.dk

We consider the vertical transport of substance in water column, modelled by advection-diffusion equation in one dimension

$$\dot{C} = -(vC - DC')'$$

where $C = C(z, t)$, $z \in [0, H]$, $t \in \mathbb{R}$, and where v and D are constant. This equation governs the concentration field of a substance which is slightly denser or lighter than water (depending on the sign of v), and subject to molecular diffusivity D .

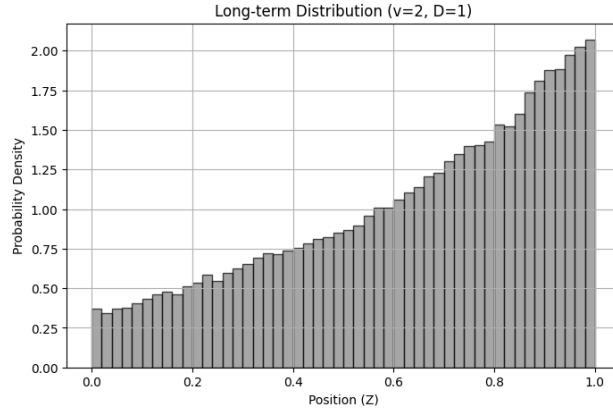
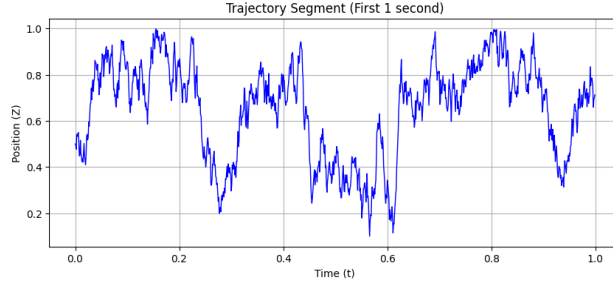
We add *no flux* boundary conditions to the equation, i.e. $vC - DC' = 0$ at $z = 0$ and $z = H$.

Question 1: Perform a Monte Carlo simulation of the transient vertical motion $\{Z_t : t \geq 0\}$ of a particle starting at the position $Z_0 = \frac{H}{2}$.

Implementation: Use reflection to implement the no-flux boundary condition: If a particle passes the lower boundary $z = 0$ during a time step, reflect it back into the domain by changing the sign of its position. Use the corresponding algorithm at the upper boundary. Take e.g. $H = 1$, $D = 1$, $v = 2$;

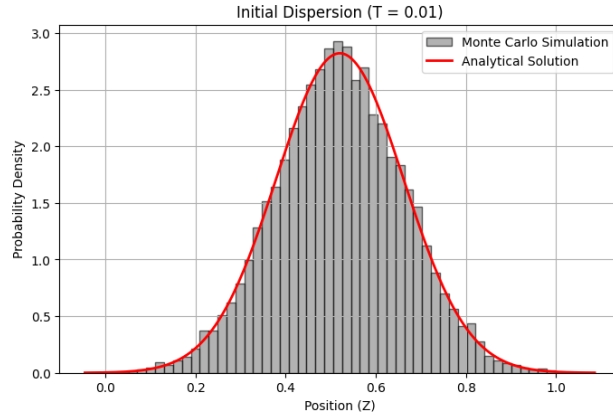
- a Investigate the long-term distribution of the particle, using one simulation of a very long duration (e.g. $T = 100$) with a moderate time step (perhaps $\Delta t = 10^{-3}$). Plot a segment of the trajectory to confirm that it looks as you would expect. Then plot the histogram of the position. Change v and repeat.

To simulate a particles movements in 1D we create a stepping function based on the given advection-diffusion equation $\dot{C} = -(vC - DC')'$. It should look like a markovian process where Z_{t+1} is entirely based on Z_t , some given velocity v , and the diffusion D . So $Z_{t+1} = Z_t + \text{velocity} + \text{diffusion}$. The velocity is given by v and should just follow that. Of course it is important to scale it by the stepping scalar, the time step, such that velocity = $v\Delta t$. For the diffusion we can write the given advection-diffusion equation as the PDE $\frac{\partial C}{\partial t} = -(vC - D\frac{\partial C}{\partial z})' = v\frac{\partial C}{\partial z} + D\frac{\partial^2 C}{\partial z^2}$. And this we can isolate diffusion by $v = 0$: $-D\frac{\partial^2 C}{\partial z^2}$. Since diffusion element should be modelled as a standard Gaussian variable $X \sim \mathcal{N}(0, 1)$ but with a variance of the given diffusion we should multiply this X with that variance V . V is the integral of the position squared (spread) times the probability density: $V = \int z^2 C(z, t) dz$. We can substitute the probability density with the diffusion and get $V = D \int z^2 \frac{\partial^2 C}{\partial z^2} dz$, which equates at $2D \int C dz$, where $\int C dz$ equals 1, since it is a probability density (the total sum of probabilities). Therefore the variance is $2D$ times the time step; $2D\Delta t$. Thereby our stepping function becomes $Z_{t+1} = Z_t + v\Delta t + \sqrt{2D\Delta t}X$. We run a simulation for $T = 100$ with $\Delta t = 10^{-3}$, and show the following trajectory segment over the first second ($T = 0 \dots T = 1$) and histogram of the distribution over the full simulation



- b Investigate the initial dispersion, using a large number of simulations of short duration (e.g. $T = 0.01$) and with short time step (e.g. $\Delta t = 10^{-4}$). Compare with the analytical solution to the advection-diffusion on the infinite domain, i.e., without boundaries at $z = 0$ and $z = H$.

We use the same stepping function with changed T and Δt parameters over 10,000 simulations, and plot the corresponding histogram



Since the velocity has only acted for $T = 0.01$ time units, the center of the cloud has barely shifted ($Z_0 + vT$, and the variance is tight around that center. We can see that since the simulation has only run for so short duration, this resembles an initial dispersion, and there is no tendency towards $Z = 1$ yet.

Question 2: Find an analytical expression for steady-state concentration, normalized so that the total amount of the substance is 1. I.e., find a solution with $\dot{C} = 0$ and $\int_0^H C(z)dz = 1$. For the parameters in question 1, plot the analytical solution on top of the histogram and compare. *Hint:* There are two possible approaches: First, write a second order ordinary differential equation that governs the concentration field and solve using standard techniques. Alternatively and easier, show that the flux $J(x)$ must vanish at each point x . Then use this to write a *first order* ordinary differential equation that governs the concentration field, and solve this using standard techniques;

$\dot{C} = 0$ means $-(vC - DC')' = 0$. Using the alternative, easier hint, we show that the flux $J(z)$ must vanish at each point z . Since the flux is $J(z) = vC - DC'$ we have that $-J'(z) = 0$. Thereby $J(z)$ must be a constant. And to find this constant we look to the boundary condition for $z = 0$, where the flux *must* be $J(0) = 0$. And since $J(z)$ is constant (because steady-state requires the flux to be constant everywhere) we know $J(z) = 0, \forall z$. Thereby we get this first order differential equation $vC - DC' = 0$. This we solve wrt. z by first isolating D and z on each side of the equation

$$\begin{aligned} vC - D \frac{dC}{dz} &= 0 \\ vC &= D \frac{dC}{dz} \\ vC dz &= D dC \\ \frac{vC}{D} dz &= dC \\ \frac{v}{D} dz &= \frac{1}{C} dC \end{aligned}$$

for which a solution is $\int \frac{v}{D} dz = \int \frac{1}{C} dC$. Since v and D are constants we say the constant $a = \frac{v}{D}$, for which $\int a dz = az \Rightarrow \frac{v}{D} z$. And as $\int \frac{1}{C} dC = \ln |C| + c$ we have that

$$\begin{aligned} \frac{v}{D} z &= \ln |C| + c \\ \frac{v}{D} z - c &= \ln |C| \\ e^{\frac{v}{D} z - c} &= |C| \\ C &= \pm e^{-c} e^{\frac{v}{D} z} \end{aligned}$$

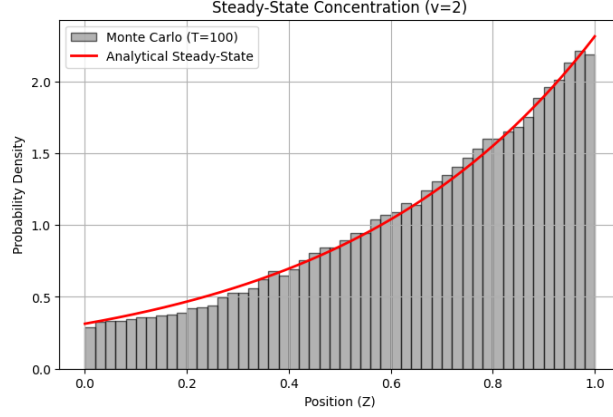
Since c is a constant e^{-c} is so too. We can therefore substitute with the constant $A = e^{-c}$. And since physical concentration cannot be negative we know A is positive

$$C = A e^{\frac{v}{D} z}$$

Combining with $\int_0^H C(z)dz = 1$ we get; $\int_0^H A e^{\frac{v}{D} z} dz = 1$, where A isolated in the above expression is $A = \frac{1}{\int_0^H e^{\frac{v}{D} z} dz}$. The integral is found as $\int e^{\frac{v}{D} z} dz = \frac{1}{\frac{v}{D}} e^{\frac{v}{D} z} = \frac{D}{v} e^{\frac{v}{D} z}$ and eventually we express $A = \frac{v}{D(e^{\frac{v}{D} H} - 1)}$ giving us a final expression for C :

$$C = \frac{v \cdot e^{\frac{v}{D} z}}{D(e^{\frac{v}{D} H} - 1)}$$

We plot the this final expression for and compare with the result from Question 1a;

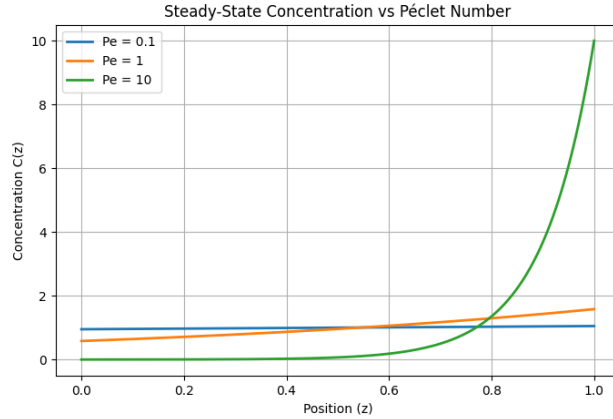


Question 3 From the parameters in the model, one may form the non-dimensional Peclet number as $Pe = |v|\frac{H}{D}$. Plot the steady-state concentration for different values of the Peclet number (for example $Pe=0.1$, $Pe=1$, and $Pe=10$). Point out how this number determines the steady-state concentration;

We rewrite the Peclet number; $Pe = |v|\frac{H}{D} \rightarrow Pe = \frac{vH}{D}$ since v is positive. And not isolate the velocity diffusion ratio; $Pe = \frac{vH}{D} \Rightarrow \frac{Pe}{H} = \frac{v}{D}$ which we can now plot into our expression for the concentration of the steady-state from Question 2;

$$C = \frac{v \cdot e^{\frac{v}{D}z}}{D(e^{\frac{v}{D}H} - 1)} \Rightarrow C = \frac{Pe}{H} \frac{e^{\frac{Pe}{H}z}}{(e^{\frac{Pe}{H}H} - 1)} = \frac{Pe}{H} \frac{e^{\frac{Pe}{H}z}}{(e^{Pe} - 1)}$$

We plot the concentration of the three different states;



The Peclet number is essentially the ratio between advection and diffusion. A low Peclet number means a diffusion dominated state where the random kicks from diffusion has far greater impact than the weak advection velocity. This means the concentration across the positions in our 1D world is almost the same, as the advection effect has effectively no impact. When the Peclet number is 1, we see a balanced state where the advection is strong enough to create a clear tendency towards the upper boundary. Lastly if the Peclet number is way beyond 1 ($Pe > 1$), the advection creates a clear tendency towards the upper boundary.

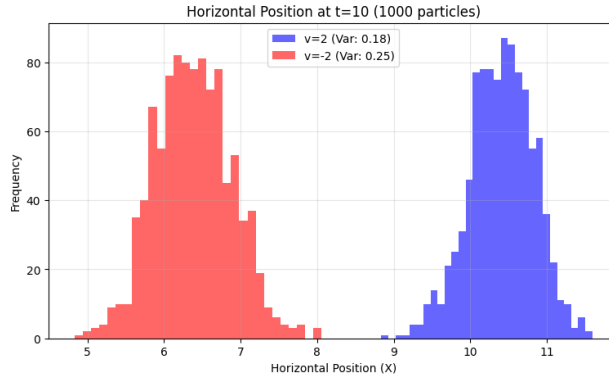
Add horizontal flow

We now add a horizontal dimension to the problem: Assume that there is a horizontal flow, so that the horizontal position $X_t : t \geq 0$ satisfies

$$\frac{d}{dt}X_t = u(Z_t) \text{ with } u(z) = \log(1 + 3z)$$

Question 4 Simulate the vertical and horizontal motion of a particle, starting at $(x, z) = (0, \frac{H}{2})$ at time 0. Take $t = 10$, simulate a trajectory up to time t , and plot. Taking $N = 1000$ replicate particles, plot a histogram of X_t for time $t = 10$, and compute the mean and the variance of X_t . Repeat for $v = 2$ and comment: Does it appear that the mean of X_t can be predicted from the vertical distribution, i.e. from the C you found in the previous? Why does the variance of X_t appear to depend on v ? If t is large enough, does it appear reasonable to disregard the vertical position and claim that the horizontal position is a random walk, i.e. that there is an effective horizontal diffusivity;

We establish the horizontal stepping function; since the horizontal equation $\frac{d}{dt}X_t = u(Z_t)$ has no diffusion term D it is a deterministic ordinary differential equation where its velocity depends solely on the particles depth Z_t . We thereby define the stepping function as $X_{t+1} = X_t + u(Z_t)\Delta t$. And simulate over $N = 1000$ particles for $T = 10$ with $\Delta t = 10^{-3}$;



We see that it is clear you can predict the mean of X_t by knowing C . A low v at $v = -2$ pulls the particles towards the lower boundary where $z = 0$, which results in a horizontal velocity of $u(0) = \log(1 + 3(0)) = 0$. The more tendency towards the upper boundary C dictates the higher horizontal position tendency. Further, if t is large enough we can practically disregard the vertical position and claim that the horizontal position is a random walk. Even though the horizontal equation is deterministic (there is no diffusivity), the randomness from the vertical axis gets translated into the horizontal variance. The accumulation of random time spent in different z points results in a Gaussian distribution for X_t with the variance of this distribution growing linearly over time.